
Input: raw audio bytes (uploaded clip)
Write to temp file
librosa.load sr=32000, mono=True (full natural-length waveform, no crop/pad)
logmel() [preprocess.py] n_fft=1024, hop=512, n_mels=64 -> librosa.power_to_db
_normalize_mel zero-mean / unit-std
Reshape to tensor (batch=1, channel=1, 64, T)
CNNBaseline.features
Conv2d 1->16, k3, pad1 BatchNorm2d -> ReLU MaxPool2d(2)
Conv2d 16->32, k3, pad1 BatchNorm2d -> ReLU MaxPool2d(2)
Conv2d 32->64, k3, pad1 BatchNorm2d -> ReLU MaxPool2d(2)
Conv2d 64->128, k3, pad1 BatchNorm2d -> ReLU
AdaptiveAvgPool2d((1,1)) -> 128 x 1 x 1
CNNBaseline.classifier
Flatten
Dropout(0.3)
Linear(128 -> 64)
ReLU
Dropout(0.3)
Linear(64 -> n_classes) (raw logits)
Softmax
Output: { scores: [...] }